Appendix E

Ion Implanter Maintenance Safety Considerations

ION IMPLANTER SAFETY CONSIDERATIONS DURING MAINTENANCE FOLLOWING ARSENIC AND PHOSPHOROUS OPERATIONS

There are a number of references dealing with handling arsenic or phosphorous residue during maintenance of the implanter.[E1][E2] The safety considerations given to arsenic or phosphorous flakes, particles, etc., are different than if one were working on arsine (AsH_3) or phosphine (PH_3) bottle exchanges.

It has been brought to our attention that there may be a mechanism where arsenic or phosphorous flakes/residue can form measurable levels of AsH_3 or PH_3 upon exposure to moist air.[E3][E4] The hydride conversion appears to be enhanced by the presence of aluminum.[E4]

Depending upon the humidity of the air which is present upon the opening of source, beamline, or end station areas, it is possible for a reaction to take place between the moisture laden air and the arsenic (or phosphorous) residues whereby small but detectable amounts of AsH_3 or PH_3 are present for short periods of time. The length of time can be several minutes to, perhaps, several hours or more depending upon the amount of arsenic or phosphorus present and the surface area involved. We recommend that access to the source housing, analyzer chamber, beamlines, and end stations of ion implanters be preceded by adding one (or more) of the following to the existing routine:

1. When venting the implanter or implanter section for planned maintenance, it is advisable to vent the sections (or system) in the usual method (generally plant N_2).
2. Once a flange, vacuum cover, or door assembly is removed or opened, the personnel should wait for five minutes prior to working in the exposed areas which need maintenance or repair. A maintenance, "scrubbed" exhaust hose should be placed in the opening if available. Use of an appropriate gas detector is advised if access to large chambers or assemblies is desired, and there has been heavy arsenic or phosphorus implanter operation (gas or solid). Use of local/state approved breathing apparatus is recommended when removing and transferring contaminated assemblies. Bagging and sealing of large assemblies is often the rule. Some sites have covered, wheeled carts which can be attached to an approved facility exhaust for storage waiting decontamination.

   Bagging is generally preferred, where feasible, to allow direct transfer to a vented space without further air exposure.[E4]
3. If a local repair is in order, such as a scanner plates adjustment or wafer size change—activities where any heavy concentration of arsenic or phosphorous are present—local scrubbed exhaust ventilation is recommended, if available, to the usual safe handling of these particular contaminated items. Again, it is recommended that an appropriate gas detector be used before access is allowed. Large assemblies from the source housing, manipulator, analyzer, or beamline areas should always be transported in a sealed environment and be stored in an appropriately vented and controlled area until properly decontaminated. Bagged parts should be unbagged immediately upon being placed in a scrubbed vent hood to minimize hydride gas accumulation in sealed bags. Used bags, gloves, wipes, etc. should remain in the vented hood for several hours until they can be placed in a vented trash receptacle, for eventual disposal in sealed hazardous waste containers.

4. If scraping and cleaning of the arsenic or phosphorous is to take place, special additional steps are required:

   a) Use a self-contained or umbilically coupled breathing system as approved by local/state standards.

   b) Use an approved HEPA filtered vacuum cleaner and place a building/facility exhaust hose (which is attached to an appropriate scrubber) at the exhaust port of the vacuum cleaner. (Note that a shroud of larger diameter duct hose may be required to fit entirely over the exhaust area, taped down to a smaller exhaust hose with cleanroom tape.) This configuration will allow the maintenance personnel to scrape and clean the heavy concentrations, all the while using the special breathing apparatus.

      When oxidized material is scraped away, new unoxidized areas could be exposed, creating additional small amounts of AsH_3 or PH_3.

   c) There is another type of vacuum cleaner (made by Nilfisk, called "Vapor Vac," which effectively captures minute amounts of toxic gases) which does not require any special venting of the vacuum exhaust since it contains a special carbon filter. Of course, the special breathing apparatus is still required.

5. In addition to venting, breathing apparatus, and vacuum cleaning issues related to cleaning arsenic and phosphorus, there is one other area of major concern with phosphorous. When there are intermediate to high concentrations of phosphorous present in the source, source housing, or analyzer chamber areas, or in parts of the source foreline, there is a risk of spontaneous combustion during scraping actions on and around the phosphorous residue. It is advised that, where possible, all parts be immersed in DI water containers under exhausted scrubbed hoods when being cleaned. The water can then be disposed of in the approved method

(check local, regional codes). Most often, it is not possible to remove an assembly for immersion. Again, use appropriate breathing apparatus and local exhaust and utilize "non-sparking" (i.e., beryllium copper) tools or Scotch-Brite™ where approved. Ensure that the appropriate fire extinguisher is nearby.

— CAUTION —
Deposit Ignition Hazard

Sparking and ignition of phosphorus residues can occur even when non-sparking tools are used. Spray all non-graphite surfaces with D.I. water before cleaning to retard ignition tendencies.

6. Ensure that all cleanwipes, etc., which are used for cleaning up phosphorus are kept in a controlled area, away from other flammable material.
7. An alternative technique for cleaning phosphorus residue is to use a wetting technique directly on the material whereby a spray bottle of D.I. water is used to "wet" the material, oxidizing it somewhat deeper than it would otherwise have oxidized naturally, in a dry state. There is some possibility that wetting arsenic and phosphorus deposits may contribute to the formation of the hydride gases, so this process should be done under a vented hood, or with scrubbed local exhaust.
    a. Spray a good portion (perhaps 2" x 6" area, 5 x 15 cm area) ensuring that the water has a chance to soak the area. Wait 10–15 seconds.
    b. Incremental spraying of DI water alternated with light scraping of the surface will form a mud-like consistency.
    c. Nilfisk Vacuum Applications personnel indicate that this material can be directly

vacuumed with their units (Models GS80, GS801, for example). The residual material within the special vacuum cleaner system should be disposed of following state/local codes as required.

Some of the techniques described in items 4 through 7 come from the procedures or suggestions from selected users. If you have any comments or suggestion of your own, please contact Varian I.I.S. Product Support (508) 281-2000 or fax directly to (508) 281-1957.

E1. Baldwin, D., King, B., Scarpace, L. Ion Implanters: Chemical and Radiation Safety, *Solid State Technology* (January 1988)

E2. Ungers, L. J. and Jones, J. H., Industrial Hygiene and Control Technology Assessment of Ion Implantation Operations, *Am. Ind. Hyg. Assoc. Journal* 47 (10) (1986)

E3. Asom, M. T., Mosovsky, J., Liebenguth, R. E.., Zilko, J. L., Transient Arsine Generation During Opening of Solid Source MBE Chambers, *Journal of Crystal Growth* 112, North-Holland (1991)

E4. Filipp, N., Hurtsaker, H., Herring, P. Investigation of Hydride Emission During The Maintenance of Ion Implantation Equipment, presented at Poster Session on June 2, 1992, American Industrial Hygiene Conference, Boston, MA (1992)